



Hives

(Urticaria; Wheals)

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Hives are red, itchy, slightly elevated swellings. The swelling is caused by the release of chemicals (such as histamine) from mast cells in the skin, which cause fluid to leak out of small blood vessels temporarily. Itching may be severe. Hives have clearly defined borders and may have a pale center. Typically, crops of hives come and go. One hive may remain for several hours, then disappear, and later, another may appear elsewhere. After the hive disappears, the skin usually looks completely normal.

(See also [Itching](#).)

Hives



Hide Details

Hives are raised, itchy, and reddish spots on the skin.

Photo provided by Thomas Habif, MD.

Angioedema

Hives may occur with [angioedema](#), which, like hives, involves swelling. However, the swelling of angioedema is under the skin rather than on its surface. Sometimes angioedema affects the face, lips, throat, tongue, and airways. It can be life threatening if the swelling interferes with breathing.

Causes of Hives

Hives and angioedema can be [allergic reactions](#).

Hives may occur when certain chemicals are inhaled, consumed, injected, or touched. These chemicals can be in the environment, foods, drugs (including medications), insects, plants, or other sources. They are harmless in most people. But if people are sensitive to them, these chemicals (called triggers or allergens) can cause an allergic reaction. That is, the immune system overreacts to the chemicals.

However, most of the time, hives are not part of an allergic reaction and an allergen (the cause of the allergic reaction) cannot be identified. For example, they may result from [autoimmune disorders](#). In

these disorders, the immune system malfunctions, misinterpreting the body's own tissues as foreign and attacking them. Also, some medications cause hives directly without triggering an allergic reaction. Some physical stimuli (such as heat, cold, pressure, friction, or sunlight) may cause hives for reasons that are not well understood.

Hives usually last less than 6 weeks and are classified as acute. If hives last more than 6 weeks, they are classified as chronic.

Acute hives

If a cause can be identified, acute hives are most commonly caused by:

- Allergic reactions (such as foods and food additives, medications, or insect bites)
- Nonallergic reactions (such as medications, physical stimuli, or autoimmune disorders)

Allergic reactions are often triggered by foods, particularly eggs, fish, shellfish, nuts, and fruits; food additives; medications; or insect bites or stings. Eating even a tiny amount of some foods can suddenly cause hives. But with other foods (such as strawberries), an allergic reaction occurs only after a larger amount is eaten. Many medications, particularly antibiotics, may cause hives. Immediate allergic reactions may also occur when a substance comes into direct contact with the skin (such as latex), after an insect bite or sting, or as a reaction to a substance that is inhaled into the lungs or through the nose.

Nonallergic causes of hives include infections, some medications, and some physical stimuli (such as pressure or cold).

In more than half of cases, a specific cause of acute hives cannot be identified.

Chronic hives

Possible identifiable causes of chronic hives are the same as those of acute hives. However, in the vast majority of cases, a cause cannot be identified (is idiopathic). Most cases of chronic hives that have no identifiable cause are thought to be due to an autoimmune reaction that itself has no detectable cause. Nevertheless, all efforts should be made to identify a cause, because elimination of a cause is the best approach to treatment.

Sometimes the cause is easily overlooked, as when people repeatedly consume a food not known to be a trigger, but the food contains a preservative or dye, or an antibiotic (such as penicillin) has been given to a cow which then is present in the milk obtained from that animal. Often, despite the best efforts, the cause remains unidentified.

Chronic hives can last for months or years, then sometimes go away for no apparent reason.

Evaluation of Hives

Not every episode of hives requires immediate evaluation by a doctor. The following information can help people decide whether a doctor's evaluation is needed and help them know what to expect during

the evaluation.

Warning signs

Certain symptoms and characteristics are cause for concern:

- Swelling of the face, lips, throat, tongue, or airways (angioedema)
- Difficulty breathing, including wheezing
- Hives that are deeply colored, that become open sores, or that persist for more than 48 hours
- Fever, swollen lymph nodes, jaundice (yellowing of the skin and eyes), weight loss, and other symptoms of a bodywide (systemic) disorder

When to see a doctor

People should call an ambulance if:

- They have difficulty breathing or are wheezing
- Their throat feels as if it is closing up

People should go to an emergency department or a doctor's office as soon as possible if

- Their symptoms are severe
- They feel progressively weak or light-headed or have severe fever or chills
- They are vomiting or have abdominal pain or diarrhea

People should see a doctor if:

- A bee sting triggers hives or swelling (to obtain advice about treatment if another bee sting occurs)
- They have other symptoms, such as fever, joint pains, weight loss, swollen lymph nodes, or night sweats
- Hives recur without exposure to a trigger
- Symptoms last for more than 2 days

If children have hives that appear suddenly, disappear quickly, and do not recur, an examination by a doctor is usually unnecessary. The cause is usually a viral infection.

What the doctor does

Doctors first ask questions about the symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause and the tests that may need to be done (see table [Some Causes and Features of Hives](#)).

Doctors ask the person to describe each episode of hives in detail and any other symptoms that occurred (such as [itching](#), difficulty breathing, or [swelling of the face and tongue](#)). They ask about the person's activities before and during the episode and about possible exposure to substances that can trigger allergic reactions, including medications or drugs being taken. The person is also asked about

specific symptoms that could suggest a cause (see table [Some Causes and Features of Hives](#)), recent infections, past allergic reactions, and recent travel.

The trigger is not always clear from the history, often because the trigger is something that may have been tolerated previously.

During the physical examination, doctors first check to see whether the lips, tongue, throat, or airways are swollen. If there is swelling, they begin treatment right away. Then doctors note how the hives look, determine which parts of the body are affected, and check for other symptoms that may help confirm the diagnosis. Doctors may use various physical stimuli to see whether they can trigger the hives. For example, they may apply light pressure, heat, or cold to the skin or stroke the skin.

People themselves should not try to trigger their hives, because a severe reaction could occur.

TABLE		
Some Causes and Features of Hives		
Cause	Common Features*	Tests
Acute hives (lasting less than 6 weeks)		
Medication and substances such as • Aspirin and other nonsteroidal anti-inflammatory drugs (NSAIDs) • Some opioids • Vancomycin • Succinylcholine (sometimes given before surgery) • Contrast agents (used with imaging tests such as computed tomography) • Possibly any medication or substance, whether prescription, over-the-counter, or herbal, if people are allergic to it	Hives can start within minutes after the drug or medication was taken and can start up to 48 hours later	A doctor's examination

Cause	Common Features*	Tests
Foods that trigger an allergic reaction (food allergens), such as peanuts, nuts, fish, shellfish, wheat, eggs, milk, and soybeans	Hives that start within minutes or hours after consumption	A doctor's examination, particularly the medical history Sometimes allergy skin prick testing
Infections (rare causes) - Bacterial, such as some urinary tract, streptococcal, and <i>Helicobacter pylori</i> infections - Parasitic, such as Toxocara canis, Giardia intestinalis (lamblia), Schistosoma mansoni, Strongyloides stercoralis, Trichuris trichiura, and <i>Blastocystis hominis</i> infections - Viral, such as hepatitis (A, B, or C) and HIV, cytomegalovirus, Epstein-Barr virus, and enterovirus infections	Fever, chills, and fatigue Symptoms of the particular infection Particularly for parasitic infections, recent travel to a developing country	Tests depending on the suspected infection (suggested by results of the medical history and examination) Diagnosis confirmed if hives disappear after the infection is eliminated
Insect bites or stings	Hives that start within seconds or minutes after an insect bite or sting	A doctor's examination
Physical stimuli - Cold - Exercise - Pressure on the skin - Scratching of the skin (urticarial dermatographism) - Heat	For most stimuli, hives that typically start within seconds or minutes of exposure to the stimulus For pressure on the skin, hives that start within 4–6 hours and affect only the area of the skin pressed	A doctor's examination Exposure to the suspected physical stimulus to see whether it triggers symptoms

Cause	Common Features*	Tests
• Sunlight • Sweating as occurs during a warm bath, exercise, or fever	For sunlight, hives that affect only the area of the skin exposed to sunlight	
Serum sickness	Hives that start within 7–10 days after injection of • A blood product (as in a transfusion) • A medication derived from animal blood such as horse serum (which is used to treat poisonous snake and spider bites) • Possibly another medication May be accompanied by fever, joint pain, swollen lymph nodes, and abdominal pain	A doctor's examination
Substances that trigger an allergic reaction through direct contact with the skin (contact allergens), such as latex or meat	Hives that start within minutes or hours after contact	A doctor's examination Sometimes allergy testing
Transfusion reactions	Hives that usually start within a few minutes after transfusion of a blood product	A doctor's examination
Chronic hives (lasting more than 6 weeks)		
Autoimmune disorders • Systemic lupus erythematosus (lupus) • Sjögren syndrome • Urticarial vasculitis	Various symptoms depending on the autoimmune disorder For systemic lupus erythematosus, symptoms may include fever, fatigue, headache, joint pain and	For all autoimmune disorders, blood tests to check for abnormal antibodies Sometimes skin biopsy For Sjögren syndrome, a test that estimates the

Cause	Common Features*	Tests
	swelling, painful breathing, and mouth sores For Sjögren syndrome, dry eyes and dry mouth For urticarial vasculitis, hives that • May be painful rather than itchy • Usually last more than 24 hours • Do not whiten (blanch) when pressure is applied • Can be accompanied by small blisters and reddish purple blotches (purpura)	amount of tears people are producing For urticarial vasculitis, skin biopsy
Cancer, typically of the digestive organs or lungs , or lymphoma	Weight loss, night sweats, abdominal pain, cough (sometimes bringing up blood), jaundice (yellowing of the skin and eyes), swollen lymph nodes, or a combination	Various tests depending on the cancer suspected
Chronic idiopathic hives (diagnosed when no specific cause is identified)	Hives that occur daily (or almost daily) and itching that lasts for at least 6 weeks, with no obvious cause	A doctor's examination Blood and sometimes other tests, such as skin prick tests and exposure to various triggers, to exclude other causes
Medications and substances (same as those for acute hives)	Hives that occur in a person who has been taking a prescription, an over-the-counter, or an herbal product for a long time when there is no other explanation for them	A doctor's examination Sometimes allergy testing Trial of avoidance to see whether hives disappear after the medication or substance is stopped

Cause	Common Features*	Tests
Endocrine abnormalities such as a thyroid disorder or an elevated level of progesterone (a female hormone)	For thyroid disorders, difficulty tolerating heat or cold, a slow or fast heart rate, and shaking (tremor) or sluggishness Occurring in women who take birth control pills (oral contraceptives) or hormone therapy containing progesterone or who have hives appearing just before their menstrual periods start and disappearing when periods stop	A doctor's examination If a thyroid disorder is suspected, a blood test to measure thyroid-stimulating hormone
Mastocytosis	Small red discolorations of the skin that turn into hives and become raised when touched Sometimes abdominal pain, easy flushing, and recurring headaches	Skin and sometimes bone marrow biopsy Sometimes blood tests to measure levels of substances released when certain immune cells (called mast cells) are activated
Physical stimuli (same as those for acute hives)	For most stimuli, hives that typically occur within seconds or minutes of exposure to the stimulus For pressure on the skin, hives that start within 4–6 hours and affect only the area of the skin pressed For sunlight, hives that affect only the area of the skin exposed to sunlight	A doctor's examination Exposure to the suspected stimulus to see whether it triggers symptoms

* Features include symptoms and results of the doctor's examination. Features mentioned are typical but not always present.

HIV = human immunodeficiency virus.

Testing

Usually, testing is not needed for a single episode of hives unless symptoms suggest a specific disorder that requires treatment (such as an infection). But if hives have unusual characteristics, recur, or persist, tests are usually done.

Typically, tests include a complete blood cell count and blood tests to measure levels of electrolytes, sugar (glucose), and thyroid-stimulating hormone and to determine how well the kidneys and liver are functioning.

Skin tests, such as a [skin prick test](#), are done by an allergist (a doctor who specializes in allergic disorders) to identify specific allergens. Imaging and other blood tests are done based on results of the history and physical examination. If results suggest that the cause is a bodywide disorder, a thorough evaluation is needed to identify the cause.

A skin [biopsy](#) is done if the diagnosis is unclear or if individual hives last more than 48 hours (though an episode of hives that come and go can last 6 weeks or longer).

Treatment of Hives

- Avoidance of triggers
- Measures to relieve itching
- Medications

Hives often go away on their own after a day or two. If the cause is obvious or if the doctor identifies a cause, people should avoid it if possible. If the cause is not obvious, people should stop taking all nonessential medications until the hives subside.

Bathing and showering with only cool water, refraining from scratching, and wearing loose clothing may help relieve symptoms.

Medications

Antihistamines taken by mouth are used for hives. These medications partially relieve the itching and reduce the swelling. To be effective, they must be taken regularly, rather than as needed. Several antihistamines, including [cetirizine](#), [diphenhydramine](#), and [loratadine](#), are available without a prescription. [Diphenhydramine](#) is an older antihistamine that is more likely to cause drowsiness than the other two. Other antihistamines include desloratidine, [fexofenadine](#), [hydroxyzine](#), and [levocetirizine](#).

Antihistamine creams and lotions are not used because they may sensitize the skin and worsen itching.

Corticosteroids taken by mouth (such as [prednisone](#)) are used if symptoms are severe and other treatments are ineffective. They are given for as short a time as possible. When taken by mouth for more than 3 to 4 weeks, corticosteroids have many, sometimes serious, side effects (see [Corticosteroids: Uses and Side Effects](#)).

Corticosteroid creams do not help.

Epinephrine narrows blood vessels and opens airways in the lungs. It is given to people who have severe reactions or [angioedema](#) who then are admitted to the hospital. People who have these severe reactions should carry a self-injecting [epinephrine](#) pen ([epinephrine](#) autoinjector) and, if a reaction occurs, use it immediately.

In about half the people with chronic hives, the hives disappear without treatment within 2 years.

[Omalizumab](#) and [dupilumab](#) are monoclonal antibodies that may be used by people whose chronic hives have continued to occur despite other treatments.

Essentials for Older Adults: Hives

Older adults are more likely to have side effects when they take the older antihistamines (such as [hydroxyzine](#) and [diphenhydramine](#)). In addition to drowsiness, older antihistamines can cause confusion and delirium and can make starting and continuing to urinate difficult. Usually, older adults should not take these antihistamines for hives.

Key Points

- Hives may or may not be an allergic reaction.
- For most people, a cause cannot be identified.
- If a cause is identified in hives that have lasted less than 6 weeks, the cause is usually an allergic reaction to a specific substance, an acute infection, or a nonallergic reaction to a specific substance.
- If hives last 6 weeks or more, the cause usually cannot be identified (is idiopathic).
- People should call an ambulance if they have difficulty breathing or if their throat feels as if it is closing up.
- People with mild symptoms should avoid any known or suspected triggers and can take antihistamines to relieve symptoms.
- People who have severe reactions should carry a self-injecting [epinephrine](#) pen and, if a reaction occurs, use it immediately.

